Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/16/2004
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Project Title:	Stress-Driven Instabilities on Helium-4 Crystals		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Comments:			
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover the mysteries of the universe obscured by Earth"s gravity		

Helium-4 is pressurized at low temperatures near 1K to grow a crystal in a rectangularly shaped chamber. After a solid seed is formed, it is slowly grown to fill the lower part of the chamber. When the crystal is grown to the height of about 5 mm, the temperature and the pressure are stabilized. The crystal is then put under an externally applied non-hydrostatic stress. The external stress is applied by the use of piezoeletric piston. The interface between the liquid melt and crystal is observed as a function of the applied stress, time and temperature. Optical interferometry techniques are employed to observe the surface profile. If the applied stress is sufficiently small, the solid height is expected simply to decrease in proportion to the square of the applied stress. Interesting instabilities are expected to occur at the interface when the applied stress is large enough to exceed a certain threshold level. The instabilities under a variety of conditions of the stress direction, crystal thickness, boundary conditions and temperature are being studied. Our objective of this project is

Task Description:	to study the surface profile of helium-4 crystals when they are placed under externally applied stress. We wish to observe the expected corrugations of surface shape and the dynamics in the surface profile developement under different applied stress conditions and to compare these observations with predictions based on the theories of stress-driven instabilities. The formation and growth of crystal shapes are fundamental problems in general. Helium crystals at low temperatures are chosed as the test material to study these important problems. Quantum crystals of helium-4 provide an excellent testing ground for the stress-driven instability effects owing to their high purity, temperature homogeneity, and rapid crystal growth over macroscopic sizes. The superfluid liquid melt provides the mechanisms for rapid heat dissipation, freezing and melting which allow observations of the dynamic processes over short time scales. We believe our studies of stress-driven instability can contribute significantly to better understanding of the problems. Understanding of stress-driven instability is also valuable for its close relation to the phenomena of practical importance such as pattern formation in epitaxial crystal growth, surface reaction rate on pre-stressed solids and surface cracking and corrosion.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A cryogenic optical interferometric apparatus for observing the surface profile of helium-4 crystal has been completed. A helium crystal growth chamber with glass windows is attached to a pumped liquid helium pot in a vacuum can. He-4 crystals are grown in the chamber by delivering pressurized He-4 gas from a source at room temperature. A He-Ne laser beam is guided into the vacuum can with a single mode optical fiber. The beam is led into a beam expander such that the emerging "beam" illuminates the 17 mm by 10 mm cross section of the growth chamber. The expanded beam is incident onto a beam splitter. One beam from the splitter is deflected and is reflected from a "reference" mirror. The other beam from the splitter passes through the crystal chamber and passes onto the "sample" mirror. The beams reflected from the reference and sample mirrors are superposed to produce interference. The superposed beam is deflected onto the bottom end of an optical fiber bundle image conduit. The fiber bundle transmits the image from the low temperature environment to room temperature section. The image at the room temperature end of the fiber bundle is recorded with a CCD camera. Preliminary testing of the operation of the interferometer has been completed. With only helium gas present in the growth chamber, an interference pattern of parallel lines corresponding to the "misalignment" of the two reflecting mirrors. After sufficient helium is admitted to fill the growth chamber with liquid and as the liquid pressure is continued to increase, the interference pattern moves uniformly in one direction. The moving pattern is consistent with a uniform increase of the index of liquid as its density is increased. When the pressure reaches the melting pressure, the appearance of a seed crystal can be clearly discerned in the interference pattern. The originally parallel background interference pattern changes to a circular pattern near the location of the seed. The initial solid typically begins to grow at one edge of the chamber where the hydrostatic pressure is presumably the greatest. As more helium is admitted into the growth chamber, the solid continues to grow and begins to spread from one to the other. The bottom surface of the chamber is eventually covered with solid. Along the perimeter of the chamber cross section, there appears to be a meniscus where the height of the solid changes rapidly. Away from the perimeter in the inner region of the cross section, the solid surface appears to be flat. The shape of this solid helium-4 is not yet understood. The temporal sequence of the crystal growth is observed and recorded with the CCD camera. After the initial filling of the bottom section of the chamber with solid, the interference pattern clearly shows the increasing height of the solid. The solid height can be seen to increase in height uniformly over the total cross section of the chamber. When helium is withdrawn from the chamber, the solid is observed to melt uniformly over the whole surface. The melting/freezing process can be controlled easily in our system. A tubular piezoelectric transducer is installed in one wall of the solid growth chamber. A piston is attached to the end of the transducer. The piston is moved by applying a high voltage (up to 5000 V) between the inner and outer walls of the piezoelectric transducer. The applied strain on helium crystals is 5x10^-8/V. The critical threshold stress is estimated to correspond to 1500 V or so. It should then be possible to span the applied stress level beyond the threshold value to observe the phenomena of stress-driven instability. Systematic studies of the effects of stress on the crystal surface are just getting under way.
Bibliography Type:	Description: (Last Updated: 01/05/2007)
Articles in Peer-reviewed Journals	Grinfeld, P. and Kojima, H. "Instability of the 2S Electron Bubbles" Physical Review Letter, vol 91, 105301-1-4 , Sep-2003
Presentation	Elkohly, T. "Interferometeric Studies of Solid He-4 under External Stress" Physics Honors Thesis Seminar, Rutgers University Apr-2004
Presentation	Kojima, H. "Crystal Growth and Shape Formation of He-4 at Low Temperatures" Seminar in Physics, Rutgers University Mar-2004
Presentation	Elkohly, T., Wada, Y., Masutomi, R., and Kojima, H. "Stress-Driven Instability on Solid He-4 Surface" 2004 NASA/JPL Workshop on Fundamental Physics in Space, Solvang, CA Apr-2004